CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 22: Trigonometrical Ratios

Every formula and result of the chapter on one sheet

1	Sine	$\sin\theta = \dfrac{\text{perpendicular}}{\text{hypotenuse}}$
2	Cosine	$\cos\theta = \dfrac{\text{base}}{\text{hypotenuse}}$
3	Tangent	$\tan\theta = \dfrac{\text{perpendicular}}{\text{base}}$
4	Cosecant	$\operatorname{cosec}\theta = \dfrac{1}{\sin\theta}$
5	Secant	$\sec\theta = \dfrac{1}{\cos\theta}$
6	Cotangent	$\cot\theta = \dfrac{1}{\tan\theta}$
7	Quotient relation	$\tan\theta = \dfrac{\sin\theta}{\cos\theta}$
8	Quotient relation	$\cot\theta = \dfrac{\cos\theta}{\sin\theta}$
9	Square relation	$\sin^2\theta + \cos^2\theta = 1$
10	Square relation	$1 + \tan^2\theta = \sec^2\theta$
11	Square relation	$1 + \cot^2\theta = \operatorname{cosec}^2\theta$
12	Sides from the hypotenuse	Perpendicular $= h\sin\theta$, base $= h\cos\theta$